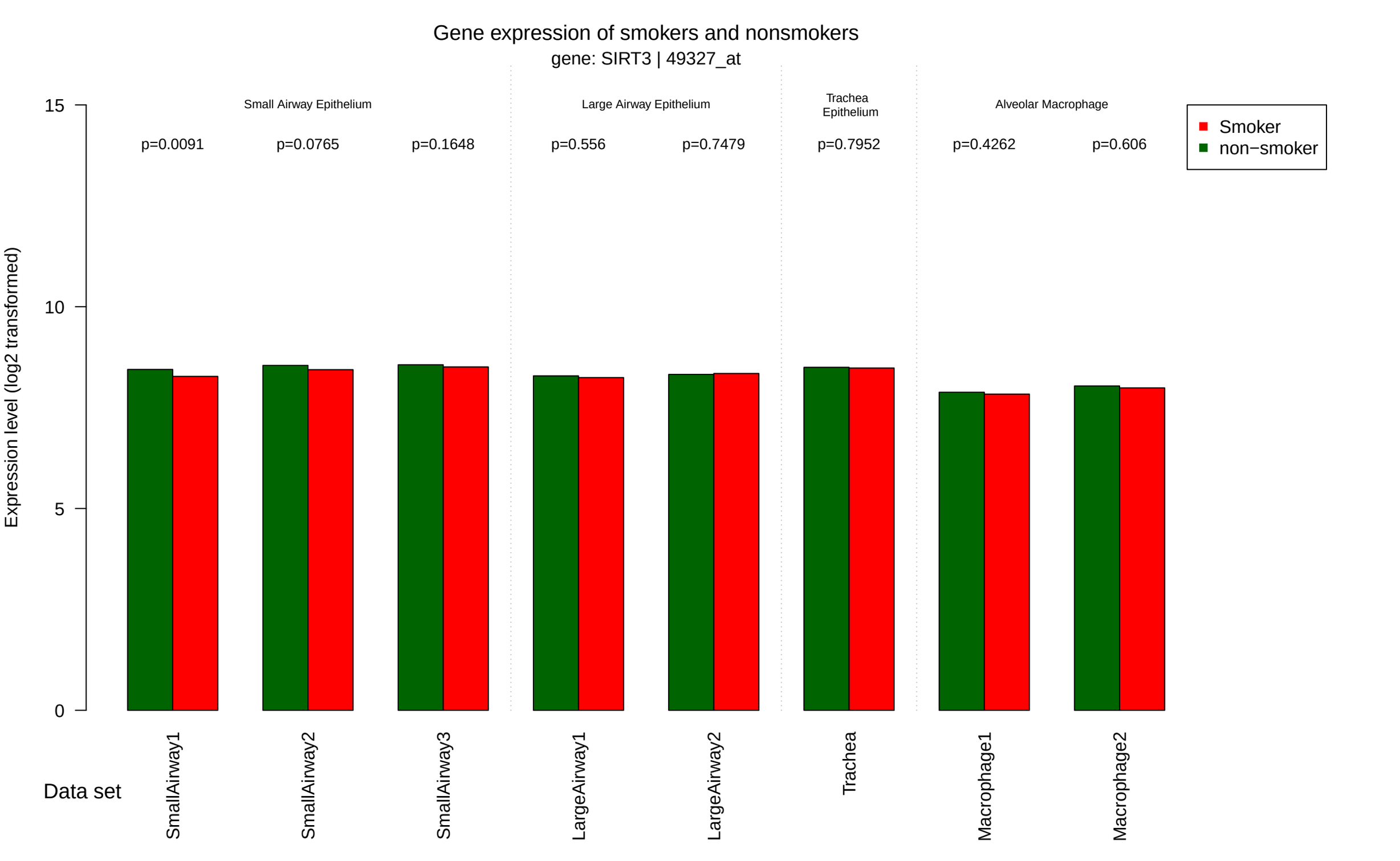
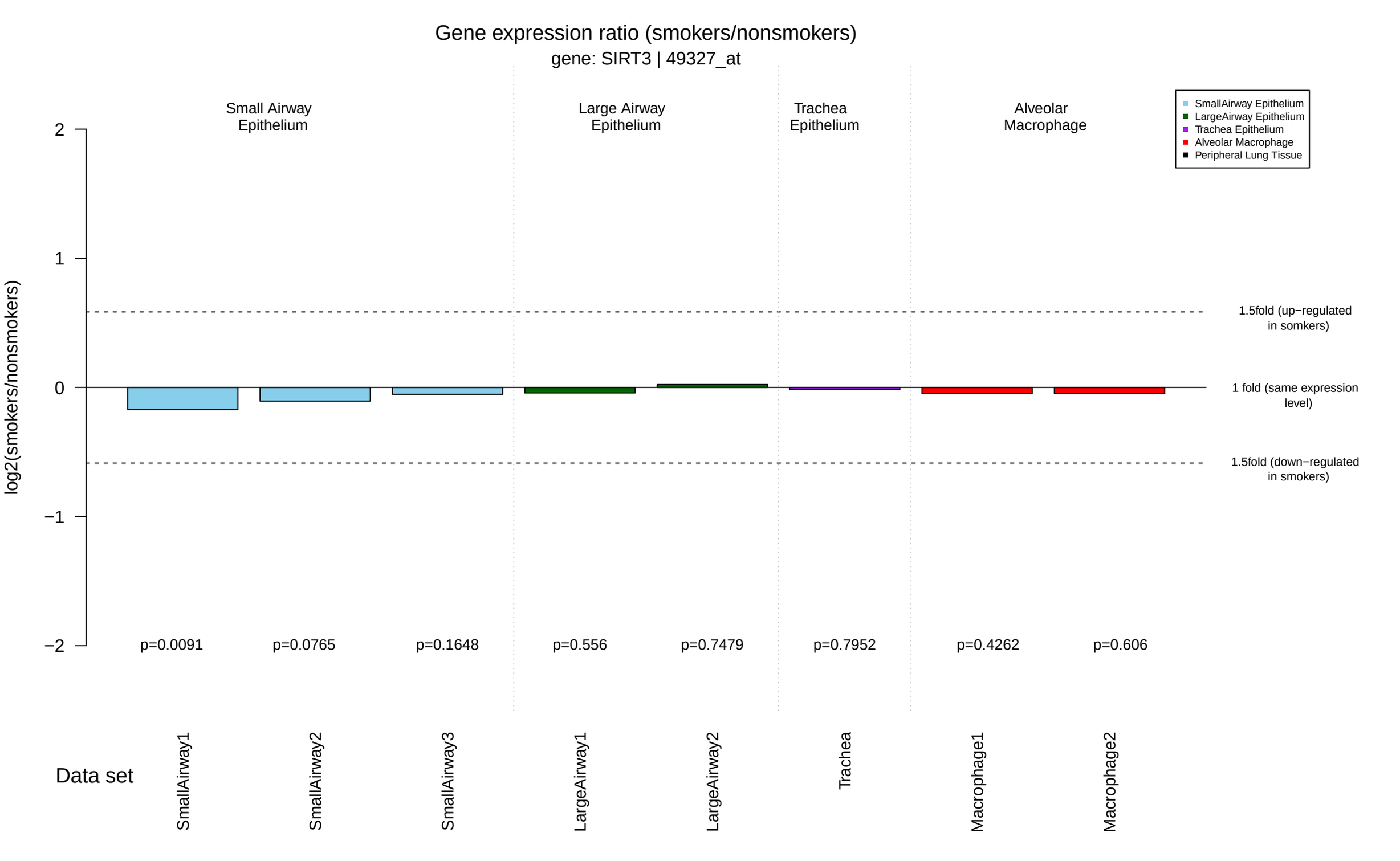
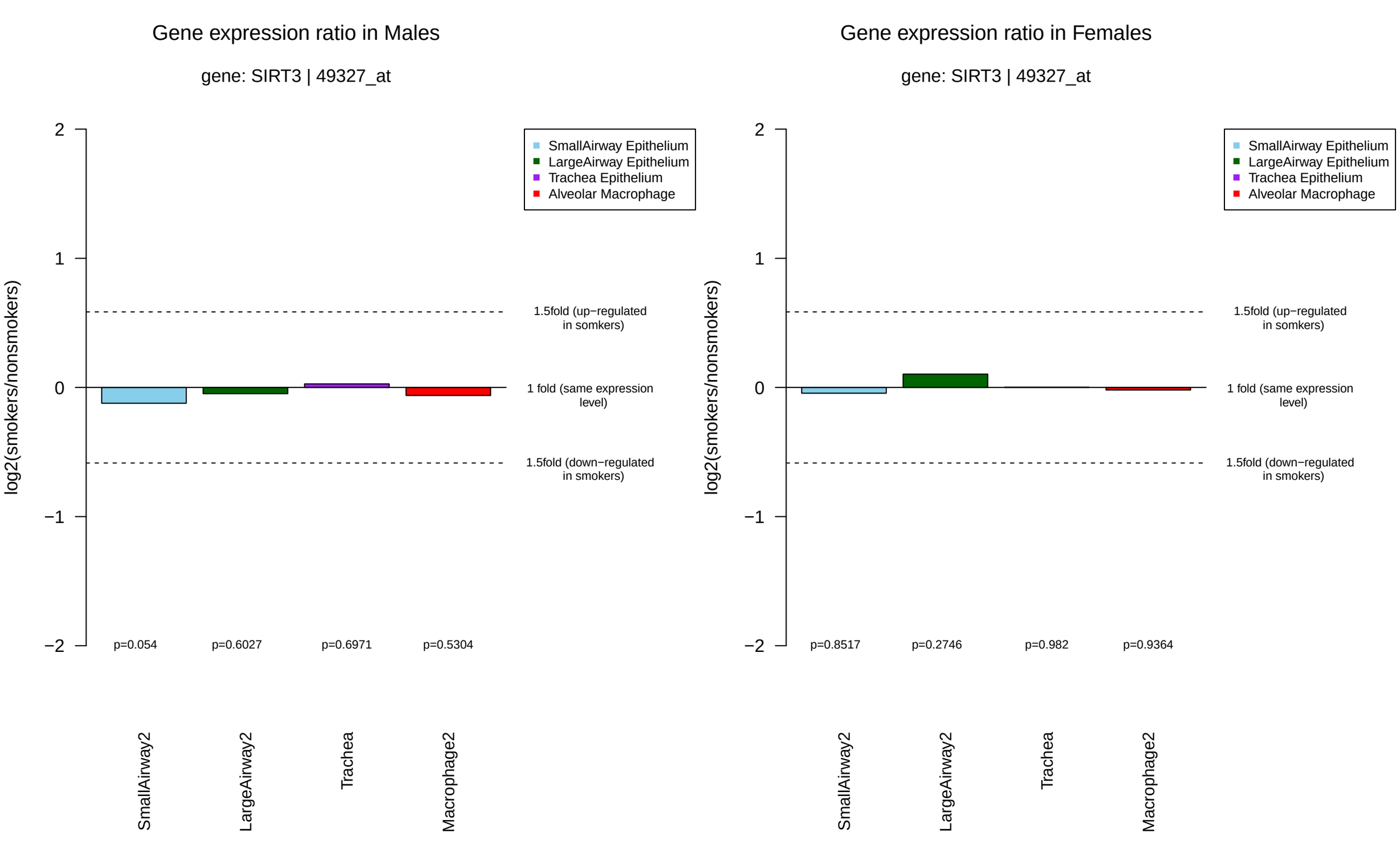
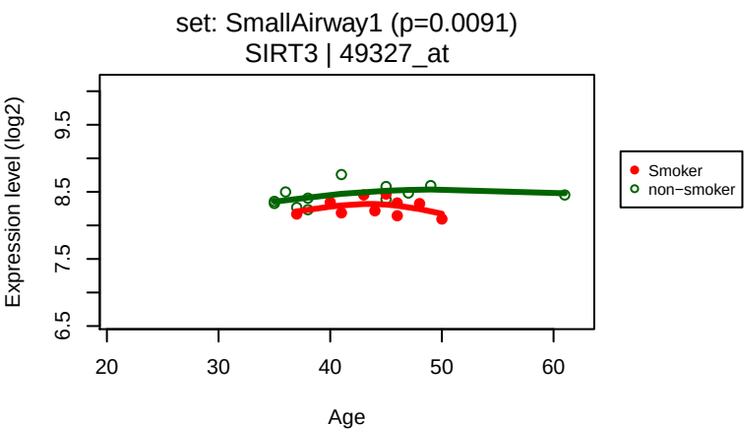
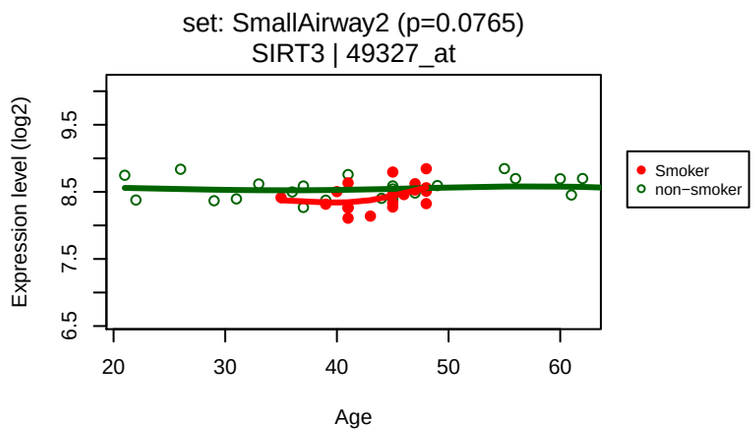
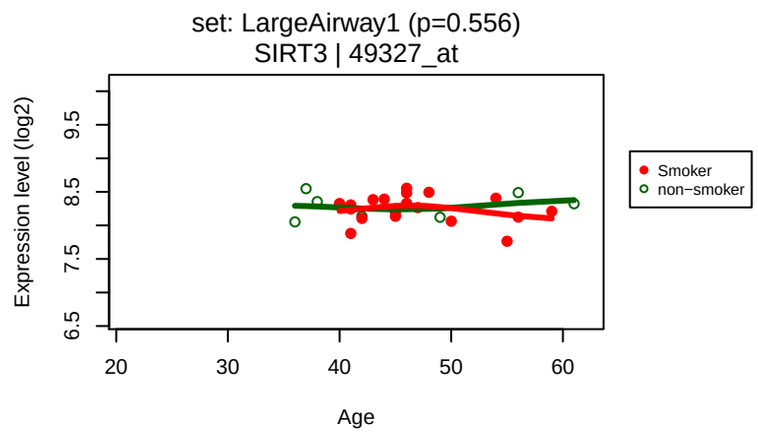

Gene expression of smokers and nonsmokers
gene: SIRT3 | 49327_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0091
p=0.0765
p=0.1648
p=0.556
p=0.7479
p=0.7952
p=0.4262
p=0.606
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

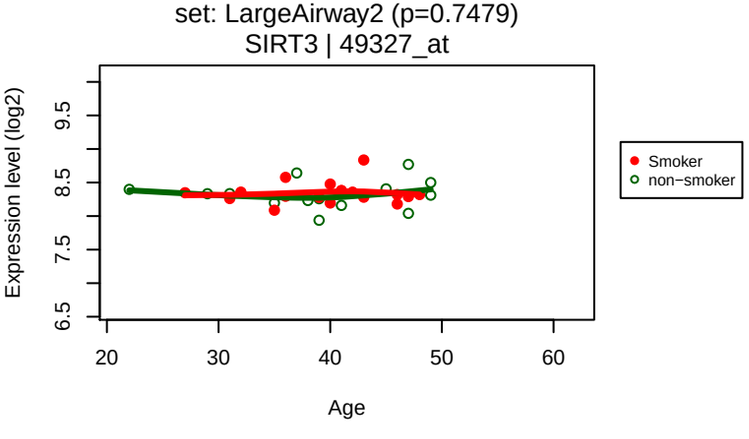

Gene expression ratio (smokers/nonsmokers)
gene: SIRT3 | 49327_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0091
p=0.0765
p=0.1648
p=0.556
p=0.7479
p=0.7952
p=0.4262
p=0.606
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

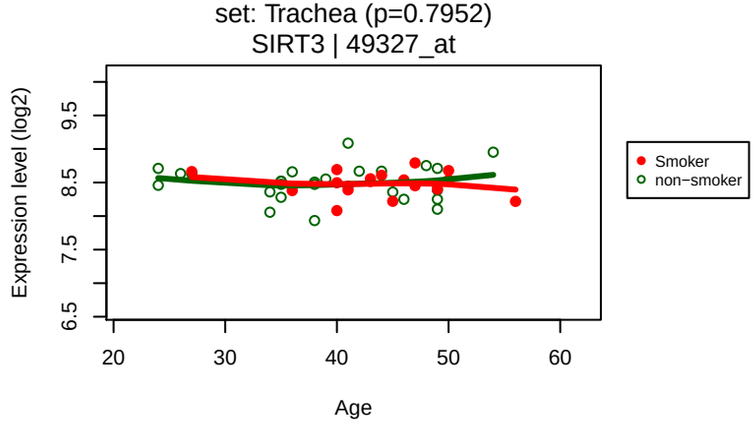

Gene expression ratio in Males
gene: SIRT3 | 49327_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.054
p=0.6027
p=0.6971
p=0.5304
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIRT3 | 49327_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8517
p=0.2746
p=0.982
p=0.9364
SmallAirway2
LargeAirway2
Trachea
Macrophage2

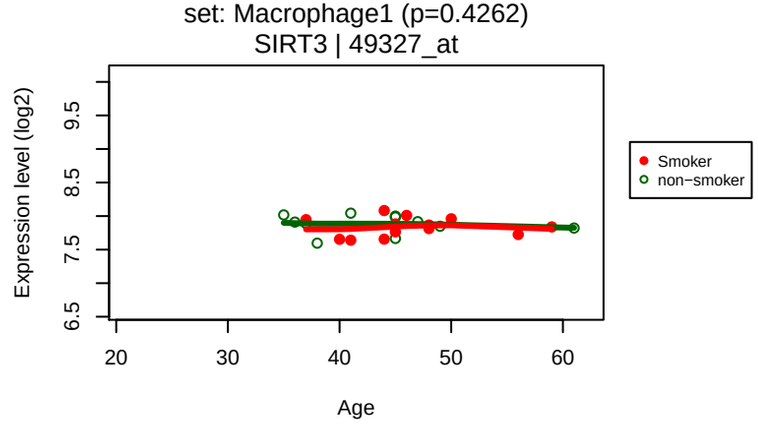

set: SmallAirway1 (p=0.0091)
SIRT3 | 49327_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker

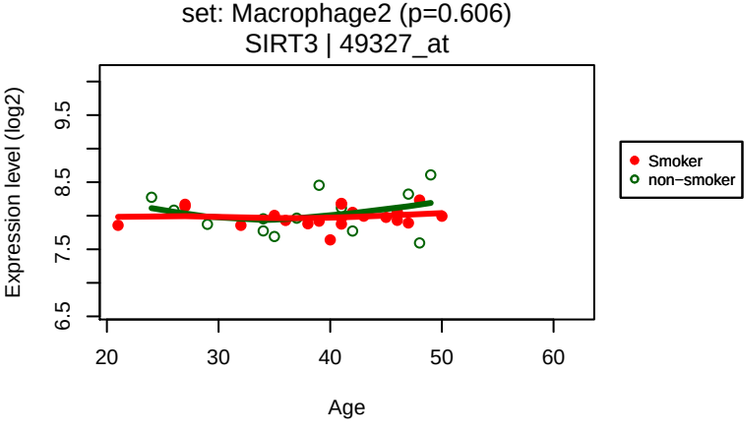

set: SmallAirway2 (p=0.0765)
SIRT3 | 49327_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker

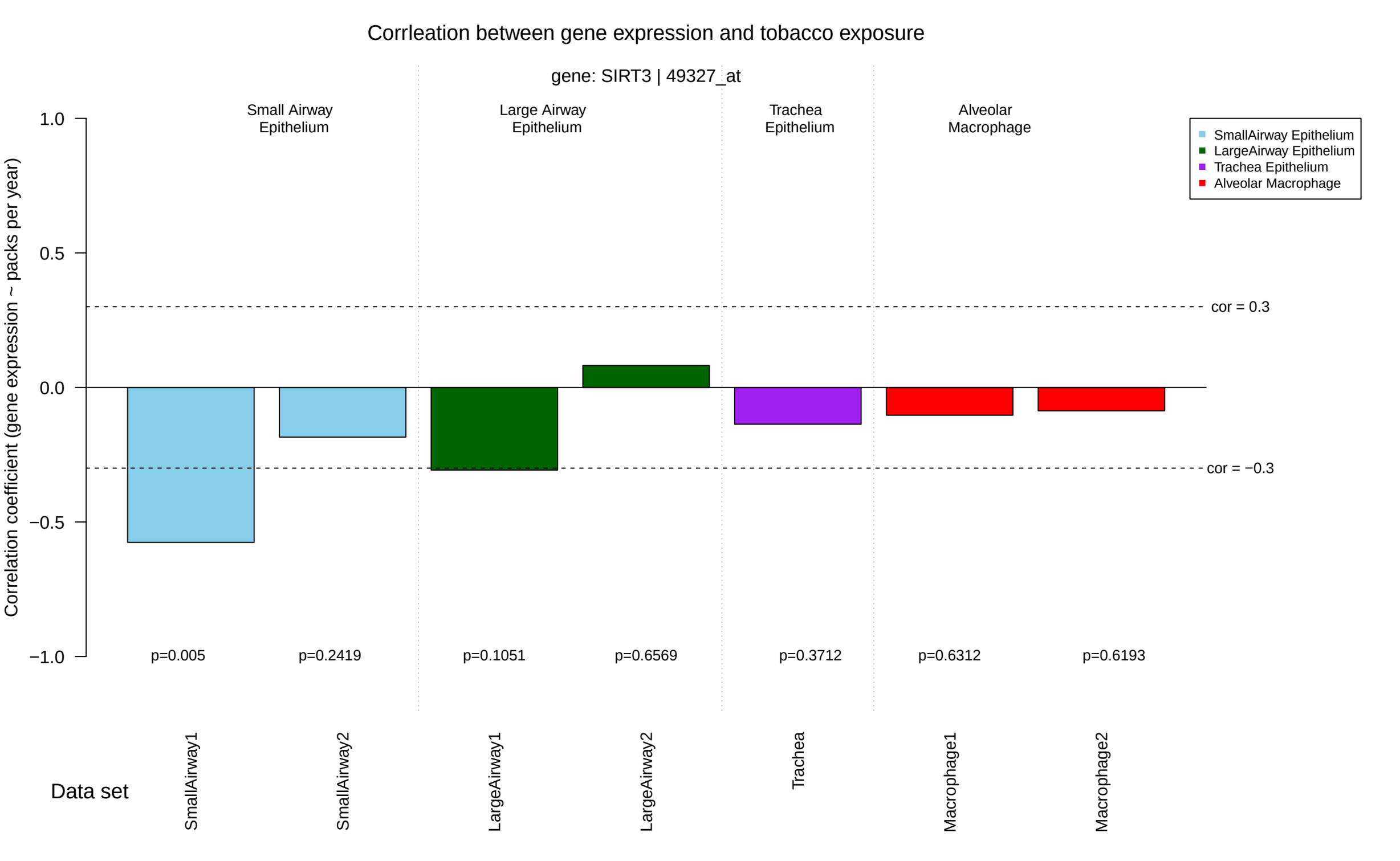

set: LargeAirway1 (p=0.556)
SIRT3 | 49327_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7479)
SIRT3 | 49327_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.7952)
SIRT3 | 49327_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4262)
SIRT3 | 49327_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.606)
SIRT3 | 49327_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIRT3 | 49327_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.005
p=0.2419
p=0.1051
p=0.6569
p=0.3712
p=0.6312
p=0.6193
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2